

Corey Heckel

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EDUCATION

Stevens Institute of Technology

Master of Science in Machine Learning, GPA: 3.97

Hoboken, NJ

May 2024 – Dec. 2024

Stevens Institute of Technology

Bachelor of Engineering in Software Engineering, GPA: 3.72

Hoboken, NJ

Sep. 2020 – May 2024

EXPERIENCE

Machine Learning Engineer II

March 2025 – Present

Nike

- Developed and deployed production recommendation models from 0 to 1, driving improvements that increased add-to-cart rate by 33%, average order value by 17%, and generated \$113M in annualized incremental demand
- Designed and deployed an AI-powered Draft Order tool for Account Executives, reducing order drafting time from 2–5 days to under 1 hour and saving 600,000+ hours of AE time annually
- Led cross-functional effort to reduce dead-end searches by 28% through embedding-based similarity matching, lifting product discovery and click-through rate by 21% and driving \$16M in incremental demand
- Migrated recommendation system serving infrastructure from Databricks to AWS, reducing operating costs by 90%, cutting latency by nearly 50% and scaling throughput from 200 RPS to 550+ RPS
- Designed and implemented a delta egress pipeline that reduced AWS egress and storage costs by 55% by replacing inefficient full table daily transfers with incremental processing

Software Engineer

Jan. 2025 – Present

Posh

- Engineered custom websites that became the primary customer acquisition channel for organizers, driving ticket sales and improving conversion rates while partnering with clients and cross-functional teams
- Integrated Google Analytics and Meta Pixel event tracking to measure user conversions and advertisement attribution

Graduate Research Assistant

May 2023 – Dec. 2024

Stevens Institute of Technology

- Built an app that empowers blind users to write using audio feedback, achieving 90% accuracy in letter recognition
- Enhanced handwriting accuracy by 30% by targeting faults in handwriting attempts determined from CNNs

PROJECTS

ForgeML | FastAPI, PostgreSQL, Redis, MLflow, Airflow, Docker, Terraform

- Architected an end-to-end MLOps platform supporting reusable training pipelines, experiment tracking, model comparison, artifact lineage, model approval, deployment readiness, and drift-triggered retraining
- Built production-style ML workflows for dataset versioning, schema validation, queued training workers, MLflow sync, model promotion, canary deployments, rollbacks, and inference health probes
- Implemented platform reliability and governance with RBAC/JWT auth, tenant isolation, audit logs, Prometheus/Grafana observability, API contracts, GitHub Actions CI, Docker, and Terraform-ready AWS infrastructure

ReelRank | PyTorch, FastAPI, LangGraph, Ollama

- Built a PyTorch two-tower recommender with metadata embeddings, ANN retrieval, reranking and FastAPI
- Developed a grounded LangGraph agent that routes natural-language prompts to explicit tools for title lookup, semantic similarity search, recommendation reranking, explanation generation, clarification, and catalog Q&A
- Implemented offline evaluation and observability dashboards tracking Recall@10, HitRate@10, NDCG@10, MRR@10, confidence intervals, train/validation loss, baseline lift, ranking attribution, and feedback signals.

TECHNICAL SKILLS

Programming Languages: Python, Go, JavaScript, TypeScript, Groovy

DB & Query Languages: DynamoDB, PostgreSQL, MongoDB, GraphQL, Redis

Data Visualization & Analysis: Pandas, Numpy, Spark, SciPy, Matplotlib, Plotly, Seaborn

AI/ML: PyTorch, TensorFlow, Keras, Sk-learn, LangChain, LangGraph, MLflow, FAISS, DCNV2, CVXPY, Hyperopt, MNRL

Cloud & MLOps: AWS (Lambda, S3, SageMaker, API Gateway, CloudWatch), Docker, Terraform, Terragrunt, Databricks, Airflow, Jenkins, CloudWatch, SignalFx, Splunk

PUBLICATIONS

Heckel, C., et al. (2026). Exploring the Effectiveness of ChatGPT in Python Code Generation. 2026 IEEE ICMI.